

An Open Source Topology Optimization Algorithm for Synchronous Reluctance Machine Design

Alexander Stewart, Nick Simpson and Phil Mellor

Abstract—Synchronous reluctance machines provide a complex high degree of freedom problem domain. Multiple optimization approaches are presented in the literature to address this with varying success and efficiency. Topology optimization algorithms show capability in efficiently optimizing designs without the need for large computational burden or compromise to the search space. The complexity of there implementation is however a large initial barrier for many. To aid in the understanding of such algorithms a level set topology optimization algorithm is presented requiring only open source platforms and minimal computational burden. The program has also been made available by the author through a open access git-hub repository.

Index Terms—Optimization, Topology, Electrical Machine, Synchronous Reluctance

I. INTRODUCTION

THE increasing demands on and applications of electrical machines throughout society has driven improvements to efficiency and power density. Often these demands are met through the use of rare earth permanent magnet topologies. Supply chain shocks in 2010 and in recent years have lead to a desire to reduce or remove such materials. The resultant method to improve a machine is thus to broaden the search of the design space to novel topologies and in doing so better tune a machine to its intended application, [3], [4].

A machine topology of particular interest is a synchronous reluctance designs. Such machines are characterized by an improved efficiency over induction machines however a poor torque quality and more complex control requirements. Several approaches are available to improve such machines however the stator and topology selection is the first to be optimized and thus critical to a successful final design. The synchronous reluctance topology problem is conventionally restricted to the rotor with a stator akin to an induction machine[13].

To broaden the design space topology optimization approaches have been proposed. Such approaches describe the optimization region as a function space to be solved rather than a parametric series of objects. In doing so the design space is significantly expanded and a high degree of freedom problem generated. This has been shown in the literature to have grate potential to improve the final design performance, [4], [11].

The complexity of the methods implemented in the literature to date is considerable. Results vary between articles and the development of bespoke solutions is often bound to individuals or institutions. The contribution of this work is to reduce the complexity of implementation of topology optimization for electrical machines, accelerating research in the field. A program will be defined and code provided for a low

computational burden open source program implementable across platforms.

The article is organized in four sections. Initially an introduction to the level set topology optimization algorithm implemented is outlined. A synchronous reluctance machine will then be presented as a problem domain. The resultant designs after application of the proposed optimization algorithm will be presented. Finally a discussion will be made into the limits of the algorithm presented and further work and areas for collaboration identified.

II. IMPLEMENTATION

Topology optimization has been applied to machines in a variety of methods the most frequent of which are SIMP, Level set and genetic algorithms, [11], [8], [7]. To avoid a reliance on high powered computing approaches reliant on large numbers of simulations are avoided. Such approaches are both energy intensive and expensive thus for sustainable lower cost research gradient based methods are preferred. SIMP and Level set optimization approaches offer a gradient based approach with comparable simulation speeds, [1], [9].

Here the level set topology optimization methodology is preferred for its deterministic boundary domains aligning better with plans for further work. The level set topology optimization algorithm developed is based on the standard mechanical compliance problem, [2]. To convert the mechanical compliance minimization scheme to an electrical machine optimization scheme a problem statement is required. A torque (τ) maximization problem is selected (1) and applied to the rotor surface (ϕ).

$$\max : f(\phi) = \tau \quad (1)$$

To solve (1) the torque developed by an electrical machine is considered (2).

$$\tau = \frac{3}{2}p(L_D - L_Q) \cdot I_D \quad (2)$$

Torque is thus a function of induction in the Direct (D) and Quadrature (Q) axis (L_D, L_Q) and current in the D axis (I_D). Induction is a measure of the magnetic co-energy (E') stored within a system for a given current. Thus by maximizing the difference in co-energy between the D,Q axis the torque is also maximized at that operating point, [8].

A. Level Set Derivation

The problem function can thus be said to be (3).

$$\begin{aligned}
max : E(\phi) &= E'_D - E'_Q \\
&= \left(\sum_{e=0}^N (X_e)^p \int B_e^D H_e^D dB - \sum_{e=0}^N (X_e)^p \int B_e^Q H_e^Q dB \right)
\end{aligned} \quad (3)$$

Where B is the flux density and H is the field strength. This has the advantage of being locally measured at individual mesh elements. Challis et al. [2], provide a detailed description of the optimization procedure. Silvesrini et al. [14] also implemented a similar energy based approach. Here a brief description of the procedure is given along with an explanation of modifications made. The level set optimization procedure is described by the first order Hamilton Jacobi equation (4).

$$\frac{\delta \phi}{\delta t} - v_n \|\delta \phi\| = 0 \quad (4)$$

Where v_n is the velocity function and t is the time step. Using the energy problem definition in (3) a maximization problem was thus be derived (5).

$$\delta \Pi = \int \left[\frac{\delta H(\phi)}{\delta \phi} (E'_D - E'_Q) + \lambda \frac{\delta H(\phi)}{\delta \phi} \right] \delta \phi \delta \Omega = 0 \quad (5)$$

$$[(E'_D - E'_Q) + \lambda] \cdot \hat{\delta}(\phi) \|\Delta \phi\| = 0 \quad (6)$$

$$H(\phi) = \begin{cases} 1, & \phi > 0 \\ 0, & \phi < 0 \end{cases} \quad (7)$$

Where ω is the surface to be integrated and H is a Heaviside function defining surface boundaries. Using the fundamental lemma for variational calculus (5) was simplified to the Euler Lagrange (6), [10]. Application of the original Hamilton Jacobi (4) to (6) allowed the Lagrange multiplier (λ) to be solved for the application of constraints. The function update scheme at the level set boundaries was then derived as (8), (9).

$$v_n = [(E'_D - E'_Q) + \lambda] \quad (8)$$

$$\delta \phi = v_n \delta t \quad (9)$$

In the simplest implementation the Lagrange multiplier will be assumed to be 0.

B. Level set implementation

The flow chart in Fig. 1 gives the level set implementation. This is broken down into three distinct sections. The level set function implementation is the section under consideration. The initial function space consist of a series of randomly distributed holes similar to [11]. Finite element evaluation of the function space is conducted through GMSH/GetDP an open source meshing / finite element platform, [6],[5].

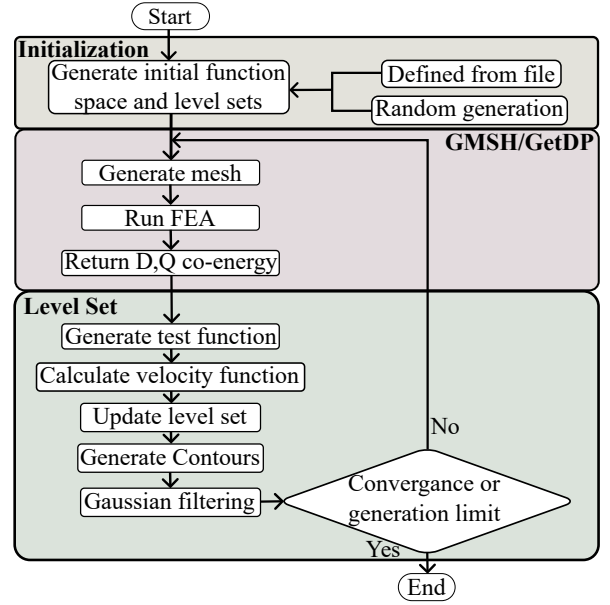


Fig. 1. Level set optimization flow chart.

C. Function space

The function space is first defined. To ensure periodic boundary regions the border of the function space is fixed to solid. This is done through setting boundary elements in the function space to the limit of the level set function (Δ). Implementation of the function space requires a relation between it and mesh elements.

The function space consists of a 3D time varying array with variables of (X,Y, ϕ). This is transferred to a level set using a contour evaluated at the zero level set. From this the model boundaries are generated and the rotor meshed.

D. Function update

The function space is updated though a test function of similar dimension (X,Y,C). Radius filtering is applied to average mesh element values for co-energy within a radius (r_{mesh}). The corresponding test function value (C) is calculated using (10).

$$C_{X,Y} = \sum \left(\frac{E'_{D_n}}{E'_{Dmax}} - \frac{E'_{Q_n}}{E'_{Qmax}} \right) \quad (10)$$

The current function space is then updated at boundary regions through a linear regression to move the boundary normal to the surface. Regression is applied to the function domain through the evaluation of values around the point under consideration of the test function. Based on the gradient of C at the boundary region the result penalizes regions with higher co-energy's in the Q than D axis.

As the density of the function space is increased boundary regions can become overly complex increasing simulation times and reducing the stability of the solution. To avoid this a gaussian filter is applied. The result is a smoothed boundary region dependent on the magnitude of the filter.

TABLE I
MACHINE PARAMETERS

Parameter	Value
Stator diameter	160 mm
Stator bore	100 mm
Poles	4
Slots per pole per phase	3
Current load	4 Amm ⁻²
Slots opening	2 mm
Slots depth	17 mm
Tooth width	6 mm
Rotor diameter	99 mm
Rotor shaft bore	30 mm

III. CASE STUDY

An example case study is proposed to highlight the efficacy of the method. A medium performance application is selected as a 5 KW pump operating at 3k RMP. The machine chosen to fit into an excising induction machine stator with the parameters given in Table I and wound with no short pitching. Induction machines in this range would be expected to operate at an efficiency of 70-90%.

TABLE II
OPTIMIZATION PARAMETERS

Parameter	Value
Time step	1 s
Δ	2
Points in X and Y	50
Gaussian filter	0.25
r_{mesh}	1 mm

A. Results

The optimization was conducted on 12th Gen Intel(R) Core(TM) i9-12900(2.4 GHz) processor with 64 GB RAM running Windows 11. The parameters chosen for optimization are given in Table II. Simulation took 22 minutes to run. Iterations were limited to 50 after which point convergence was seen.

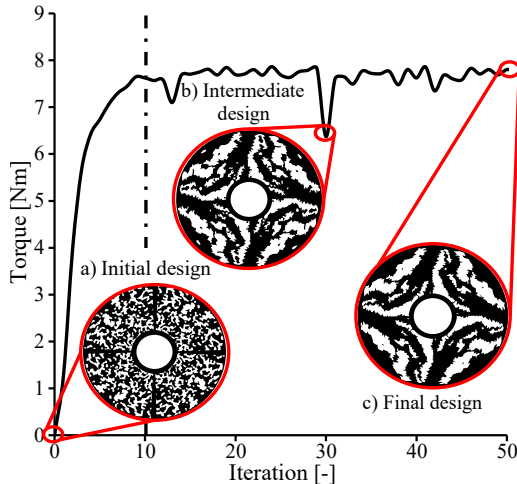


Fig. 2. Output torque of designs during optimization. Three designs highlighted : a) initial design, b) intermediate poor design and c) final design.

The designs output torque as the design progressed is optimized is shown in Fig. 2. From the initial design the program takes 10 iterations to converge towards a design. Further iterations smooth boundary features and remove smaller components from the design. Several drops in performance are seen, notably at 12 and 30 iterations.

The optimization recovers from these dips opposing the previous change in boundary. This highlights a challenge in synchronous reluctance machine design where small changes to features can lead to large impacts on performance. In the case of iteration 30 (design b)) a central barrier was temporally broken increasing Q axis flux leakage degrading performance. The final design was considered after 50 iterations.

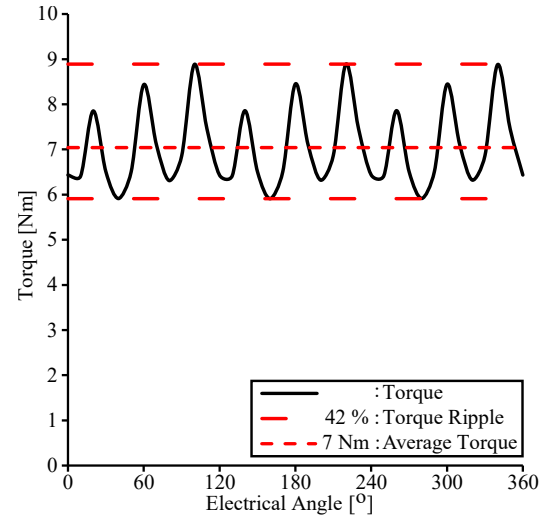


Fig. 3. Final design c) output torque. average torque of 7 Nm, and torque ripple of 42%.

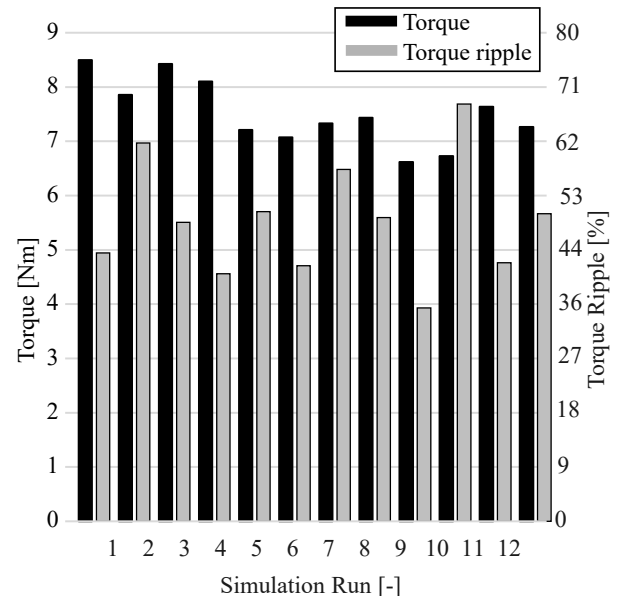


Fig. 4. Distribution of output torque. Average torque of 7.5 Nm and average torque ripple of 51.4 %.

The design shown in Fig. 2 by design c) and features two flux barriers. Several small anomalies are still present in the design and four radial bars can be seen. These bars are a result of the optimization requirement for material at pole boundaries. Convergence is assumed at this point to reduce optimization times. The final output torque of the machine is shown in Fig. 3. The design achieved a 42% torque ripple with a 7 Nm average torque. The contributions to torque ripple are associated with slot number (31%) and phase number (12%).

B. Discussion

The methodology presented generates an initial random distribution. Level set techniques are gradient based and thus influenced by the initial condition. The result of this is that when implemented it is preferential to run several iterations before drawing conclusions. The intended result being a guide from which to further optimize from. When this is done over 12 iterations the resultant torque and torque ripple distributions are seen in Fig. 4. The results achieved an average torque of 7.5 Nm with a torque ripple of 51.4%.

Applying statistical measures to the results further analysis was conducted. The torque has a standard deviation of 0.6 Nm and torque ripple has a standard deviation of 9.6 %. This variance in results is inherent to level set topologies do to local nature of the search. These standard deviation may be minimized by better tuning the optimization algorithm or further considerations over the initial condition.

The topology presented in Fig. 2 is characterized by a high torque and high torque ripple. To improve the feasibility of the topology a constraint added to reduce the torque ripple. In this example it will to limit the variation in torque ripple ($\delta \tau$) to bellow 10% (11). The algorithm was considered to be improved if the average torque ripple result lay beyond on standard deviation of the average presented in Fig. 2 or bellow 41.9%.

$$\min, \delta \tau \quad (11)$$

C. Torque Ripple Constraint

Machine rotation was found to be a poor analogy for torque ripple in the literature, [12]. work conducted by the author backed this up, thus torque ripple was directly measured. It is of note that the assumption that torque ripple is a variation in the energy stored within the rotor as the machine rotates was found to be a poor analogy for torque ripple in the literature, , thus torque was directly measured. This is done through a maxwell stress tensor through the air gap boundary. The torque ripple in Fig. 3 was considered to select an appropriate sample. 30% of the torque ripple in Fig. 3 is attributed to stator slots.

It is thus proposed that evaluating the one period of the slotting ripple is adequate to significantly reduce torque ripple. A Lagrangian (12) is thus proposed to minimize the measured torque ripple.

$$\lambda = \nabla \gamma \phi_{best} \frac{\delta \tau_{aim} - \delta \tau_{current}}{\delta \tau_{best}} \quad (12)$$

Where γ is a wind up function, ϕ_{best} is the lowest torque ripple function space result, τ_{best} is the lowest torque ripple result

, τ_{aim} is the aim torque ripple, $\tau_{current}$ is the current torque ripple and ∇ is the sign of the gradient between the current and previous torque ripple. An optimization problem was thus again set up. Parameters used in Table I were maintained along with Table II. The number of iterations was however extended to 100 let the torque ripple converge. Torque was measured at 45 degrees to the D axis and recorded at a sample of 5.6 degrees electrical four times. The wind up function was selected as (13)

$$\gamma = \begin{cases} 0, & \text{iterations} \leq 10 \\ \gamma + 0.01, & \gamma < 0.25 \\ 0.25, & \gamma \geq 0.25 \end{cases} \quad (13)$$

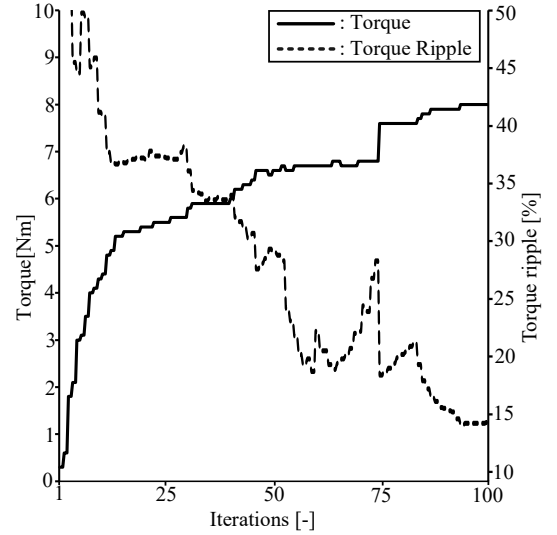


Fig. 5. Output torque and estimated torque ripple of designs during optimization.

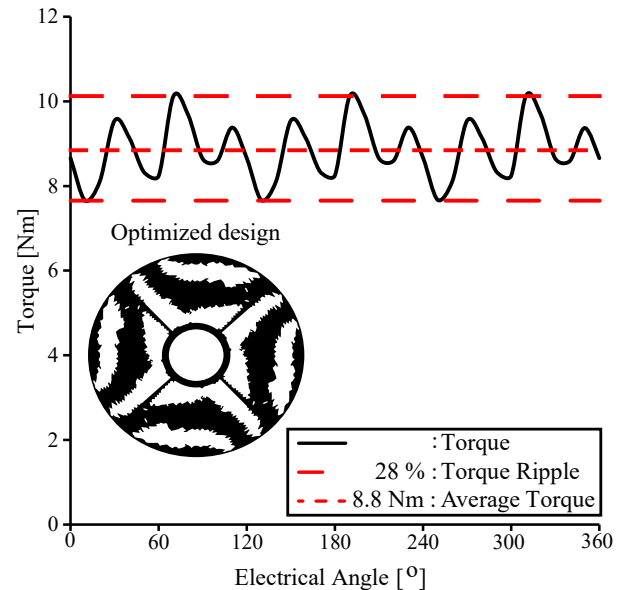


Fig. 6. Output torque of ripple optimized design. Expected ripple doubled due to small sample rate

The result optimization progression is seen in Fig. 5. The results show improvements to both torque and torque ripple as the optimization runs. Several spikes in both torque ripple and torque are seen similar to Fig. 2 again highlighting the care when optimizing synchronous reluctance machines. The optimization algorithm correctly responds to this converging towards a measured low torque ripple high torque design.

When the torque was observed a increase in torque ripple was seen compared to that measured during optimization. This was due to the small sample taken during the optimization process. Limiting the extent to which torque can be observed. When Fig. 3 and Fig. 6 are compared a shift in the ripple angle was seen making efficient measurement of the torque ripple challenging.

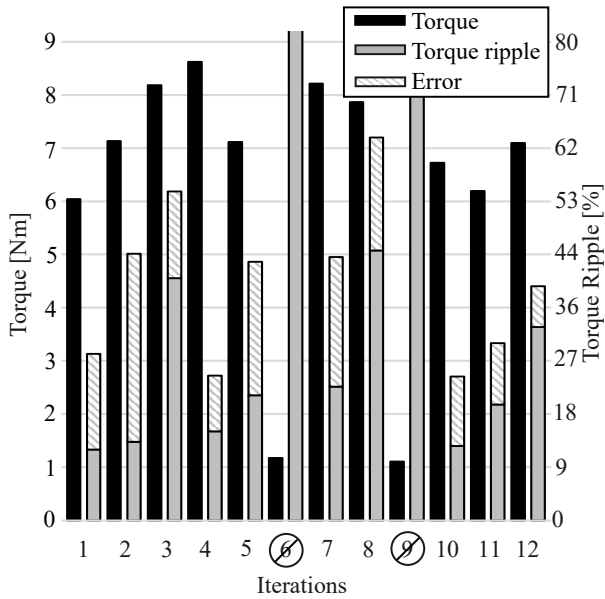


Fig. 7. Distribution of output torque. Average torque of 7.3 Nm and average torque ripple of 22.8 % / 38.8 %. Iteration 6,9 failed to converge.

Evaluation of torque ripple and increasing the number of iterations results in a increase in optimization time to 1 hr 15 min. This was considered acceptable when compared to results presented in similar optimization approaches, [15], [1]. The simulations were again run over 12 iterations and the results presented in Fig. 7. Both the estimated torque ripple during optimization and error in torque ripple between estimate and actual are presented.

Iterations six and nine are excluded from the average as they failed to successfully converge. The resultant average torque was 7.3 Nm with a standard deviation of 8.8 Nm. This suggests no meaningful change to the motors torque producing ability however a poorer convergence towards local optima. The torque ripple shows a improvement in average torque ripple by 12.6% with a increase in standard deviation. Torque ripple can thus be said to be reduced however at the cost of the stability of the solver. Results six and nine show a failed convergence suggesting the need for further tuning to the algorithm.

IV. CONCLUSION

A Open source topology optimization algorithm has been derived and implemented in python and GMSH/GetDP. Using the energy difference method for a synchronous reluctance topology initially the design was optimized to maximize torque. The algorithm was shown capable to achieve results within 30 minutes. When 12 optimization runs were compared an average torque of 7.5 Nm and a 50% torque ripple was found from an initial random condition.

To address the large torque ripple a constraint was added through a Lagrangian multiplier and assessment of torque ripple. The modifications extended the optimization time to 1 hr 15 minutes. When compared against the initial setup the modified algorithm showed no significant change to average torque while reducing average torque ripple by 12.5%. However an increase to the standard deviation of results was seen after implementation of the constraint suggesting the need for further optimization of the run parameters.

Further work is planned to present the implementation of a mechanical domain within the FEA solver. In doing so a multiphysics evaluation of the design domain can be conducted and thus mechanical constraints added. The implementation of manufacturing degradation into the optimization algorithm is also planned.

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